



AdaptiveRoute

Operational route intelligence

Agentic replanning

A fine-tuned routing policy for speed.
A MILP solver for the guarantee.

Gabriel Carvalho Domingos da Conceição

Routes are planned once. Reality changes all day.

A delivery route is optimised the night before. By mid-morning it is already wrong.

NETWORK

A road closes

An accident blocks the segment the driver was routed through.

DEMAND

A customer cannot receive

Store shut, nobody on site, delivery window missed.

PRIORITY

The order changes

A delivery becomes urgent after the plan was frozen.

Today the driver phones an operator, who re-plans by hand. **The optimisation asset built overnight is abandoned by 10am.**

Two obvious approaches, each with a disqualifying flaw

OPTION A

Ask a language model

Speaks the driver's language. Answers in seconds.

But: no notion of capacity. It will route straight through the blocked road.

OPTION B

Re-run the optimiser

Provably correct. Respects every constraint.

But: it needs a formal model of the disruption, not a sentence.

The gap is not optimisation, and not language. **It is the handoff.**

The model proposes. The solver decides.

The fine-tuned model is treated as a **fast candidate generator**, never as an authority. Every plan it produces is parsed, checked against deterministic rules, repaired if it can be repaired safely, and replaced by the MILP solver if it cannot.

Nothing reaches the driver without passing the same validation gate — **regardless of who produced it.**

Breadth where the problem is open. Precision where it is narrow.

ORCHESTRATOR · OFF THE SHELF

Qwen2.5-Coder-32B

Reads the driver's message, extracts the operational event, answers questions about the route.

general purpose · temperature 0.1

ROUTING POLICY · FINE-TUNED

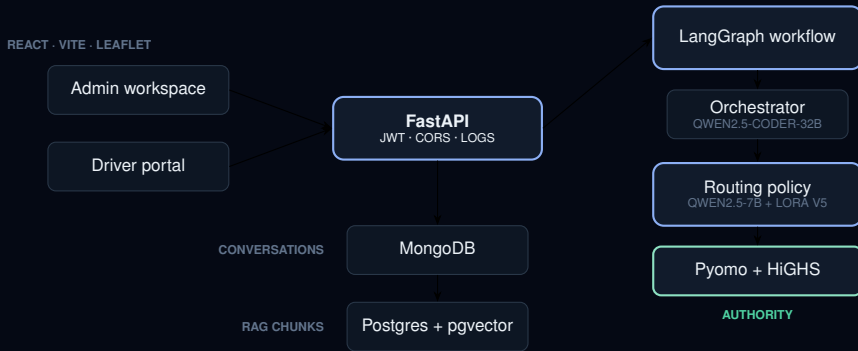
Qwen2.5-7B + LoRA v5

Emits the sequence of stops. Nothing else.
Trained on the solver's own output.

specialised · temperature 0

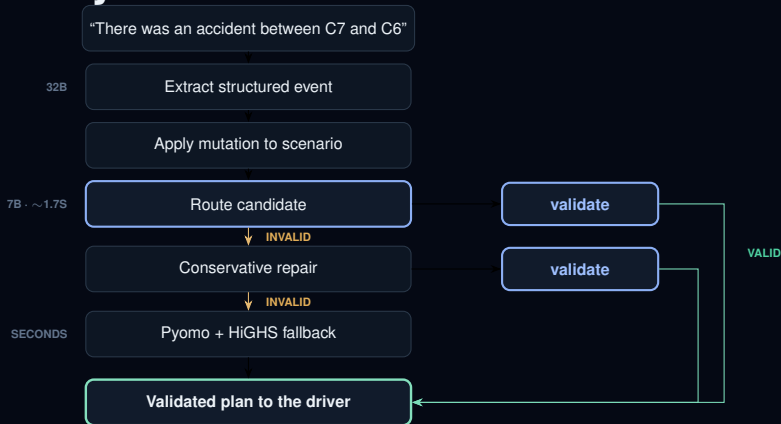
The 32B never touches a route. **The 7B never talks to a driver.**

FastAPI core, pluggable everywhere it matters



Engine, policy and event extractor are swappable by environment variable. Tests run in-memory — no GPU, no database, no model server.

Each stage is slower and more trustworthy than the one above it



When the existing plan survives the event, it is reused and the model is never called.

One arbiter, independent of who wrote the plan

A PLAN IS ACCEPTED ONLY IF EVERY RULE HOLDS

- Every active customer visited **exactly once**
- Each route starts and ends at the depot
- Vehicle capacity respected
- No blocked arc traversed
- Only known vehicles and active customers
- No duplicate or inactive stops

No exceptions. The same gate applies to the model, the repair and the solver.

What makes the conversation work

MEMORY

Context window

A bounded summary per conversation: last event, last plan, open constraints.

The driver never repeats themselves.

RETRIEVAL

Documentation in pgvector

Project docs chunked and embedded; cosine search returns supporting passages.

Answers cite sources, not recall.

GROUNDING

Facts before the model

Route facts are computed from the stored plan, then handed to the model.

The answer cannot drift from the plan.

Questions are answered from the stored plan. **Only disruptions enter the replanning workflow.**

Admin console

The screenshot shows the AdaptiveRoute Admin Console dashboard. The top navigation bar includes the AdaptiveRoute logo and the text 'Admin Console'. The main dashboard is titled 'ADAPTIVEROUTE - DASHBOARD' and 'Routes of the day'. It features a sidebar with navigation links: Dashboard, Route Chat, Scenarios, Drivers, and Knowledge. The main content area displays a map of New York City with a selected driver route (ROUTE-001) for NDAH PATEL. The route is shown as a blue line with stops labeled D0, C1, C2, C3, C4, C5, C6, C7, C8, C9, and C10. The total distance for the route is 25.30. To the right of the map, there is a 'Scenario selector' section with a search bar and a list of scenarios: 'Otm 02 - 10 orders' (1 route(s) - 10 stops) and 'Demo CVRP Scenario' (2 route(s) - 8 stops). Below the scenario selector is a 'Driver route selector' section with a search bar and a list of routes: 'ROUTE-001' (NDAH PATEL - 1000-010 - assigned) with a total distance of 25.30. At the bottom right, there is a 'Latest Agent Route Event' section titled 'Incident snapshot' which shows 'No active disruption' and a button to 'Open chat'.

Real road routing

Geometry snapped to the street network by OSRM, not straight lines between stops.

Fleet and scenarios

Orders ingested, scenarios persisted, routes assigned to named drivers.

Disruption watch

The incident panel is where a driver's report surfaces on the operations side.

Everything so far, running

ADAPTIVEROUTE · CRAT

Route replanning chat

Talk to the route agent when a driver reports blocks, delays or delivery constraints.

ROUTE CONVERSATIONS History

- For route ROUTE-002, ...
ROUTE-002
20.4k mps, 22:51
- For route ROUTE-002, ...
ROUTE-002
20.4k mps, 22:46
- For route ROUTE-002, ...
ROUTE-002
20.4k mps, 22:42
- For route ROUTE-002, ...
ROUTE-002
20.4k mps, 22:42
- For route ROUTE-002, T...
ROUTE-002
20.4k mps, 22:37
- For route ROUTE-002, T...
ROUTE-002
20.4k mps, 22:37

ADAPTIVEROUTE AGENT

Route operations chat

While naturally, The agent extracts the route id, event and constraints from the message.

ROUTE VISUALIZATION

ROUTE-002

Vehicle V2

For route ROUTE-002, the third stop is C3. The full sequence for your assigned route is: D0 → C4 → C5 → C3 → C1 → D0. The total distance for this route is 9.31 route-distance units.

User

For route ROUTE-002, What is my fourth-stop and what is the full sequence for my assigned route?

AdaptiveRoute Agent

For route ROUTE-002, the fourth stop is C1. The full sequence for your assigned route is: D0 → C4 → C5 → C3 → C1 → D0. The total distance for this route is 9.31 route-distance units.

User

For route ROUTE-002, I am delayed by 25 minutes. Which delivery should I prioritize first?

AdaptiveRoute Agent

For route ROUTE-002, the first delivery stop is C4. You should prioritize delivering to C4 first. The total distance for the route is 9.31 route-distance units.

CONVERSATION HISTORY

Agent execution

- route_lookup
- route_fact_builder
- route_qp_req
- route_qp_model

PLAN OUTPUT

Generated route

Route	Total distance	Validation
1	9.31	—

V2
4 stops - loaded 20

D0 → C4 → C5 → C3 → C1 → D0

ROLLING MEMORY

Context window

Summary
No context window loaded yet.

Refresh context

Refresh messages

ADAPTIVEROUTE AGENT

Admin Console

Logout

Detailed route: ROUTE-002

Tell the route agent what happened. Example: Replan ROUTE-001. C1 to C3 is blocked because of an accident.

Send

Follow-up

Agent execution

The trace from the validation slide, step by step: lookup, fact builder, retrieval, model.

Validated plan

Route, total distance and validation state — the contract, enforced.

Rolling memory

Context window carrying facts and constraints across the conversation.

Supervised fine-tuning where the optimiser is the teacher

1. Generate a synthetic CVRP instance
2. Solve it to optimality with HiGHS
3. Inject an operational event
4. **Re-solve** the disrupted scenario
5. Validate, then serialise the pair

The solver labels its own training set.

Four variants, one fixed benchmark

1,000 held-out scenarios, identical across every run.

	TRAINING DATA	FEASIBLE	CAPACITY	BLOCKED ARC	EXACT
v3	30K compact curriculum	91.7%	72	11	4.6%
v4	+ 70K hard continuation	92.9%	61	10	5.8%
v5	+ 20K error-driven	94.4%	46	9	5.6%
v6	+ 10K further error-driven	94.0%	50	10	5.4%

v5 selected. Highest feasibility, fewest violations of both kinds, 100% valid JSON everywhere. v6 confirmed the curve had flattened.

Where the model stops working

Measured, not estimated.

CUSTOMERS	VALID	FAILURE MODE
8	100%	—
9	100%	—
10	60%	customer omitted
12	33%	customer omitted
14	0%	customer omitted

Small samples: 3–5 scenarios per size.

THE BOUNDARY

8–9 customers

Above it the model omits customers — it does not route them badly.

A cardinality limit, not a routing one.

What we would fix before anyone depends on this

SCALE

MTZ relaxation is weak. Solver calls still run inside the request.

DATA

Fully synthetic. No real traffic, telemetry or road network.

THE UNGUARDED EDGE

Validation covers the plan, not the question

It proves the plan is right *for the mutated scenario* — not that the mutation matches what the driver said.

This is the next thing to close.

In priority order

1. Harden event extraction
2. Push the cardinality boundary
3. Move optimisation off the request path
4. Instrument the cascade in production
5. Validate against real operations

The system stays correct when the model is wrong

- The routing policy returns **94.4% feasible replans** at the size it was trained for
- The remaining **5.6% is harmless** — validation, repair, fallback
- A measured boundary of **8–9 customers**, not an assumed one
- Natural language to validated route, **end to end**

The interesting result is not the model — it is everything built around it.

Where this sits in the literature

EXACT ROUTING

Dantzig, Ramser 1959

The truck dispatching problem.
Management Science 6(1).

Miller, Tucker, Zemlin 1960

Integer programming formulation of
TSP. *J. ACM* 7(4).

Huangfu, Hall 2018

Parallelizing the dual revised
simplex method. *Math. Prog.
Comp.* 10(1).

LEARNING FOR OPTIMISATION

**Vinyals, Fortunato,
Jaitly** 2015

Pointer networks. *NeurIPS*.

Kool, van Hoof, Welling 2019

Attention, learn to solve routing
problems! *ICLR*.

Bengio, Lodi, Prouvost 2021

ML for combinatorial optimization: a
tour d'horizon. *EJOR* 290(2).

EFFICIENT LLM ADAPTATION

Hu et al. 2021

LoRA: low-rank adaptation of large
language models.
arXiv:2106.09685.

Dettmers et al. 2023

QLoRA: efficient finetuning of
quantized LLMs. arXiv:2305.14314.

Prior work replaces the solver. **This keeps it as the authority.**

Training configuration

ADAPTER

Base model	Qwen2.5-7B-Instruct
Method	LoRA / QLoRA
Rank r	16
Alpha	32
Dropout	0.05
Quantisation	4-bit NF4
Compute dtype	bfloat16
Epochs	1

RUNS — EACH CONTINUES THE PREVIOUS ADAPTER

	ROWS	STEPS	LR	TIME
v3	24,000	3,000	5e-5	5h53m
v4	56,000	2,300	1.7e-5	3h44m
v5	16,000	1,500	1e-5	2h22m
v6	8,000	800	7e-6	1h18m
total				~13h

One epoch per run, batch 8, cosine schedule, RTX 3090.

Only the routing policy is fine-tuned. The orchestrator stays off the shelf.